

Song Search System.

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Project Overview

01

**Schema
refinement**

02

**Semantic
analysis**

03

Interface

04

**Data
Evaluation**

05

Queries

06

Conclusion



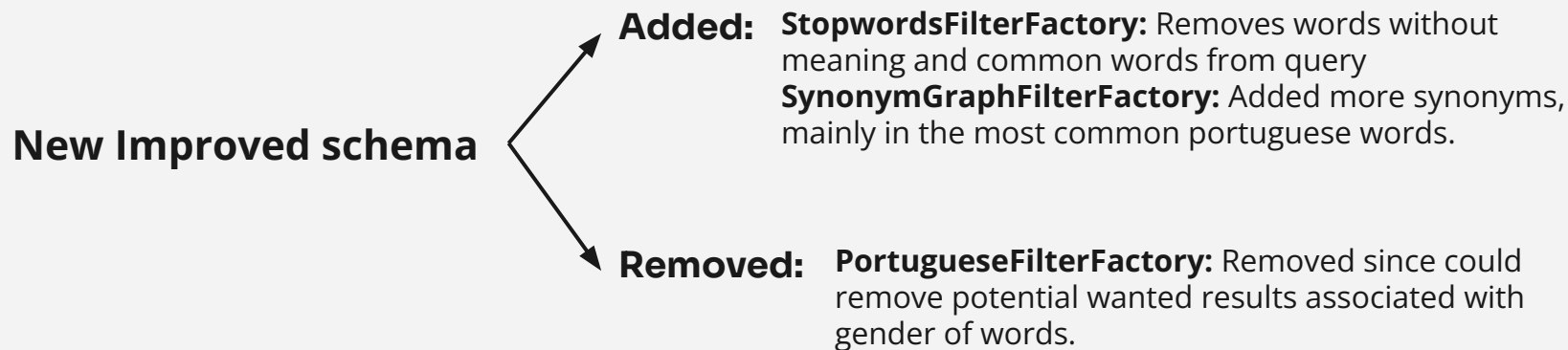
Milestone 2 Overview

- Initial search system
 - Data Indexing
 - Simple schema with default data types
 - Advanced schema with synonyms, filtering and stemming
 - Data Retrieval
 - eDisMax with qf, pf and ps
 - Data Evaluation

Semantic analysis

- **Generate document embeddings:**
 - Model used:
neuralmind/bert-base-portuguese-cased
- **Added new solr schema with dense vector**
- **Adapt previous queries to accept dense vectors**

Schema refinement



Interface

- Simple
- For showcase purposes, not designed for a potential user.
- Allows search non-semantic and semantic.

Song Search System

Non-Semantic ▾

pop e pexoval

Search

Results:

root: 2668232c-839c-4a65-b281-59fd4438550b

album: Outras Histórias

artist: Deolinda

features: {}

lastfm_content_artist: Deolinda is a project from Lisbon, Portugal, inspired by fado and its traditional roots in order to create original songs based on portuguese traditional and popular music.

It came to life in 2006, by the hands of 4 young musicians with diverse musical backgrounds and experiences (such as classical music, jazz, ethnical and traditional music), searching to recreate a sound rooted in popular music through the crossing of different musical languages and musical research serving as a base for the group's original compositions. Their first album - Canção ao Lado was released in 2006.

"... Her name is Deolinda and she is old enough to realize that life isnt as easy as it seems. Merrily unmarried, in love and out of love, born in Lisbon, she inhabits a ground floor apartment somewhere in the suburbs of the capital. She writes her own songs by peeking through the curtains of her window, drawing inspiration from the old gramophone records that once belonged to her grandmother and by the bizarre and strange life of her neighbours. She lives with 2 cats and a gold fish. "

Taken from the website <http://www.deolinda.com.pt/>

Deolinda are: Ana Bacalhau (vocals), Zé Pedro Leitão (doublebass), Pedro da Silva Martins (classical guitar) and Luis José Martins (classical guitar). Read more on Last.fm User contributed text is available under the Creative Commons By-SA License; additional terms may apply

lastfm_different_listeners: 733

lastfm_image_album: <https://lastfm.freetls.fastly.net/u/300x300/2e601d9b54b43a2e94dc5cd920419be.jpg>

lastfm_image_artist: <https://lastfm.freetls.fastly.net/u/300x300/2a96cbd8b46e442fc41c2b86b621562f.png>

lastfm_playcount: 2459

Data Evaluation

Used Metrics

Recall: The percentage of pertinent papers that were recovered from the whole collection of pertinent documents.

Precision: The percentage of pertinent documents returned.

Precision at 10 and 30 (P@10 & P@30): A targeted precision statistic that assesses how relevant the top 10/30 documents that were retrieved are.

Average Precision (AvP): Highlights the ranking order of pertinent things by calculating the average precision values at each time a relevant document is obtained.

Mean Average Precision (MAP): The mean accuracy of all the evaluation's queries.

Query

Songs with lyrics related to religion.

Query 1 - Semantic

```
"query": "{!knn f=vector topK=30 }(religião)",  
"fields": "*,score",  
"params": {  
  "start": "0",  
  "rows": "30"  
}
```

Query 1 - Improved

```
"query": "religião",  
"fields": "*,score",  
"params": {  
  "defType": "edismax",  
  "q.op": "AND",  
  "sort": "score desc",  
  "start": "0",  
  "rows": "30",  
  "qf": "lyrics^4 title^2 lastfm_content_artist^3 tag^10 artist",  
  "pf": "lyrics^6 lastfm_content_artist^6 title^10",  
  "ps": "3"  
}
```

Query 1 - Results

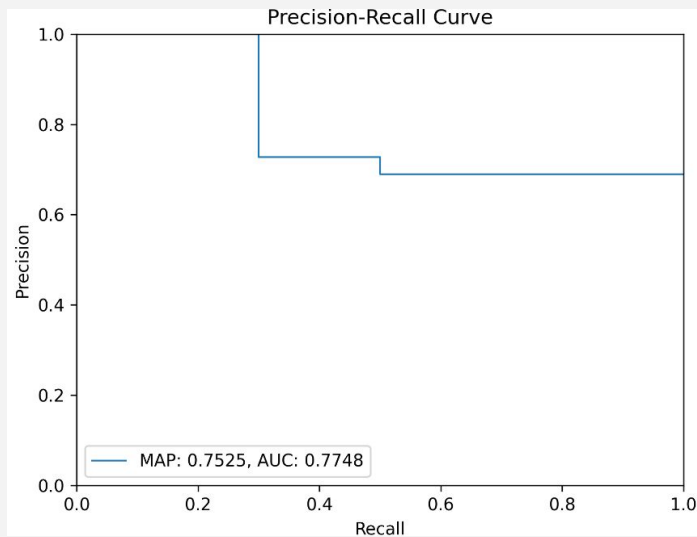
Rank Systems	Simple	Advanced	Improved	Semantic
P@10	0.5000	0.6000	0.7000	0.0000
P@30	0.3333	0.7333	0.6667	0.0333
AvP	0.5272	0.7317	0.7525	0.0417

Table 9: P@10, P@30, and AvP for query 1

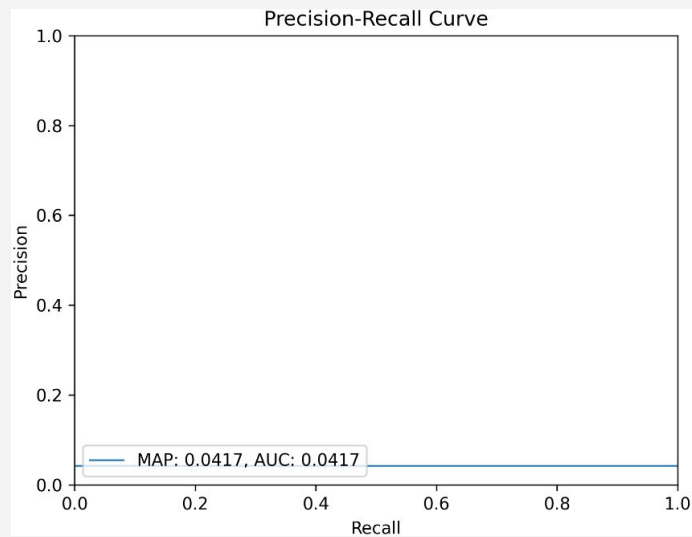
Query

Songs with lyrics related to religion.

Query 1 - Improved Schema



Query 1- Semantic Schema



Query

Songs whose artist has hit number 1 in the charts and is Portuguese.

Query 2 - Semantic

```
"query": "{!knn f=vector topK=30 }(number 1 AND portuguese)",
"fields": "*",
"score": "score",
"params": {
  "start": "0",
  "rows": "30"
}
```

Query 2 - Improved

```
"query": "number 1 AND portuguese",
"fields": "*",
"score": "score",
"params": {
  "defType": "edismax",
  "q.op": "AND",
  "sort": "score desc",
  "start": "0",
  "rows": "30",
  "qf": "lyrics^4 title^2 lastfm_content_artist^3 tag^10 artist",
  "pf": "lyrics^6 lastfm_content_artist^6 title^10",
  "ps": "3"
}
```

Query 2 - Results

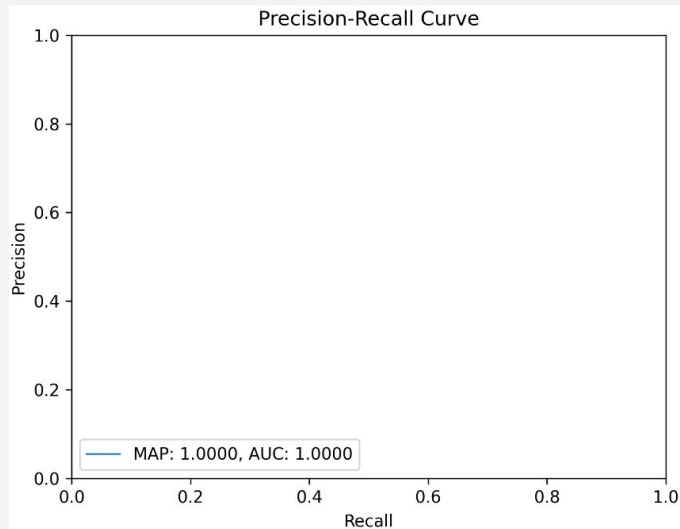
Rank System	Simple	Advanced	Improved	Semantic
P@10	0.3000	1.0000	1.0000	0.0000
P@30	0.1333	0.6667	0.6667	0.0333
AvP	0.3000	1.0000	1.0000	0.0417

Table 10: P@10, P@30, AvP for query 2

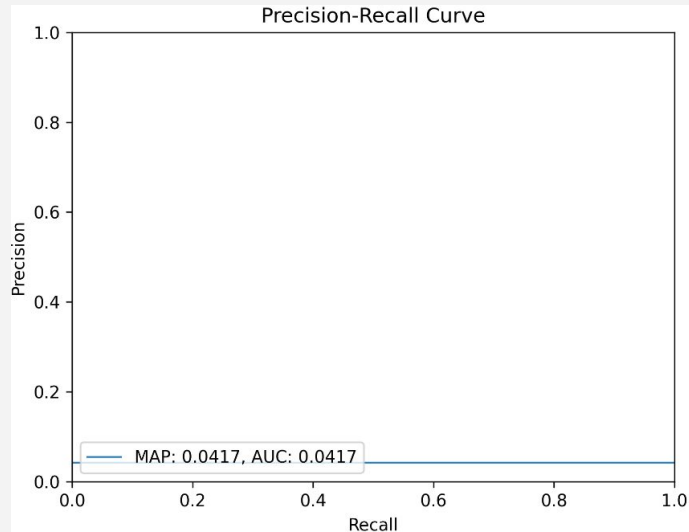
Query

Songs whose artist has hit number 1 in the charts and is Portuguese.

Query 2 - Improved Schema



Query 2- Semantic Schema



Query

Songs whose artist is female that talk about empowerment.

Query 3 - Semantic

```
"query": "{!knn f=vector topK=30 }(female AND poder)",  
"fields": "*,score",  
"params": {  
  "start": "0",  
  "rows": "30"  
}
```

Query 3 - Improved

```
"query": "female AND poder",  
"fields": "*,score",  
"params": {  
  "defType": "edismax",  
  "q.op": "AND",  
  "sort": "score desc",  
  "start": "0",  
  "rows": "30",  
  "qf": "lyrics^4 title^2 lastfm_content_artist^3 tag^10 artist",  
  "pf": "lyrics^6 lastfm_content_artist^6 title^10",  
  "ps": "3"  
}
```

Query 3 - Results

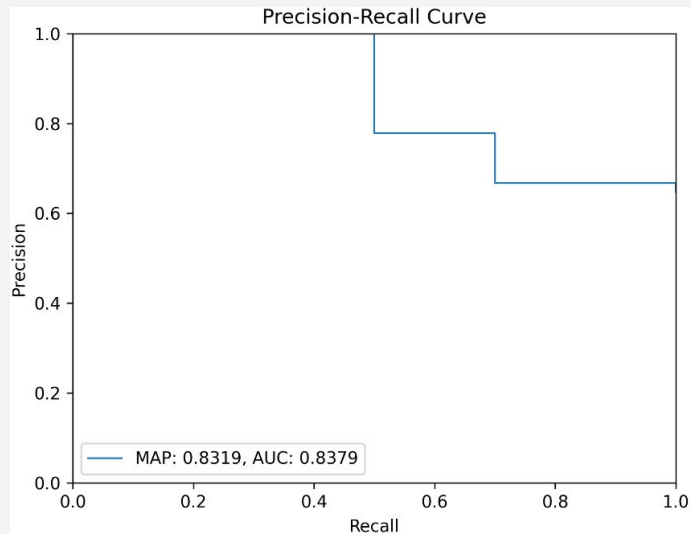
Rank System	Simple	Advanced	Improved	Semantic
P@10	0.1000	0.8000	0.7000	0.1000
P@30	0.0667	0.7333	0.3667	0.1000
AvP	0.1534	0.8501	0.8319	0.2550

Table 11: P@10, P@30, AvP for query 3

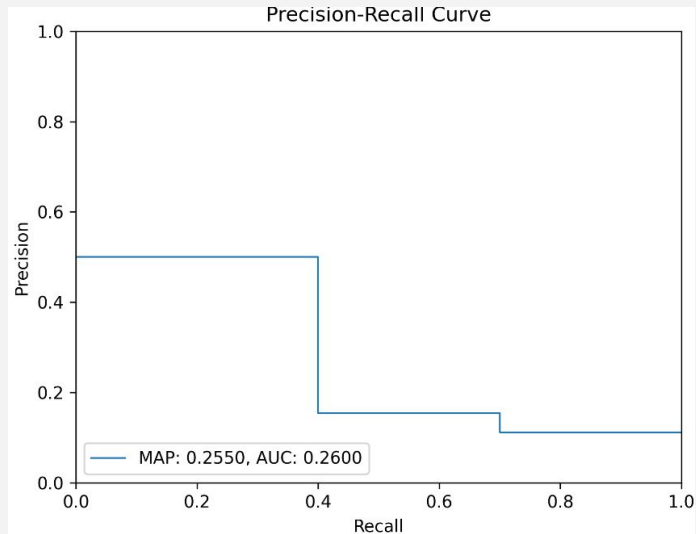
Query

Songs whose artist is female that talk about empowerment.

Query 3 - Improved Schema



Query 3- Semantic Schema



Query

Songs that talk about reflection or love.

Query 4 - Semantic

```
"query": "{!knn f=vector topK=30 }(reflexão OR amor)",  
"fields": "*,score",  
"params": {  
  "start": "0",  
  "rows": "30"  
}
```

Query 4 - Improved

```
"query": "reflexão OR amor",  
"fields": "*,score",  
"params": {  
  "defType": "edismax",  
  "q.op": "AND",  
  "sort": "score desc",  
  "start": "0",  
  "rows": "30",  
  "qf": "lyrics^4 title^2 lastfm_content_artist^3 tag^10 artist",  
  "pf": "lyrics^6 lastfm_content_artist^6 title^10",  
  "ps": "3"  
}
```

Query 4 - Results

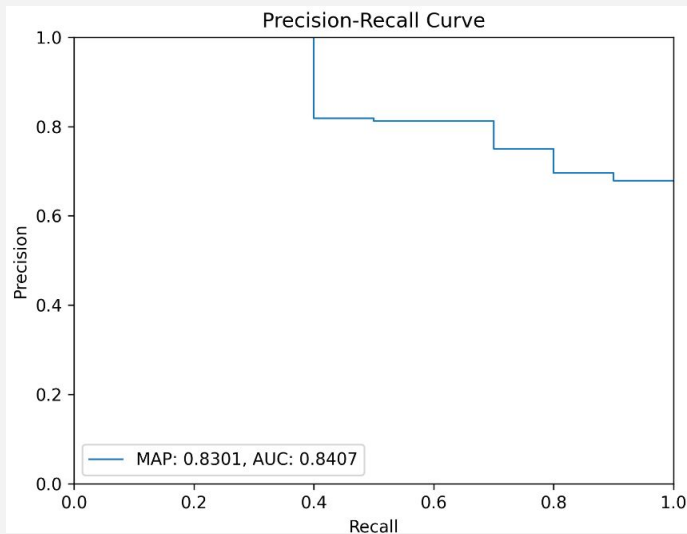
Rank System	Simple	Advanced	Improved	Semantic
P@10	0.6000	0.8000	0.8000	0.7000
P@30	0.6333	0.7333	0.6333	0.5667
AvP	0.7326	0.8087	0.8301	0.6840

Table 12: P@10, P@30, AvP for query 4

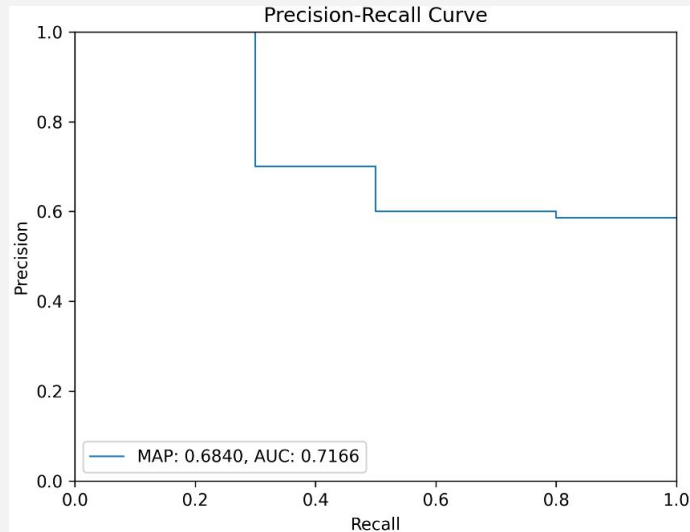
Query

Songs that talk about reflection or love.

Query 4 - Improved Schema



Query 4- Semantic Schema



Query

Pop songs that talk about the past.

Query 5 - Semantic

```
"query": "{!knn f=vector topK=30 }(pop AND passado)",  
"fields": "*",  
"params": {  
  "start": "0",  
  "rows": "30"  
}
```

Query 5 - Improved

```
"query": "pop AND passado",  
"fields": "*",  
"params": {  
  "defType": "edismax",  
  "q.op": "AND",  
  "sort": "score desc",  
  "start": "0",  
  "rows": "30",  
  "qf": "lyrics^4 title^2 lastfm_content_artist^3 tag^10 artist",  
  "pf": "lyrics^6 lastfm_content_artist^6 title^10",  
  "ps": "3"  
}
```

Query 5 - Results

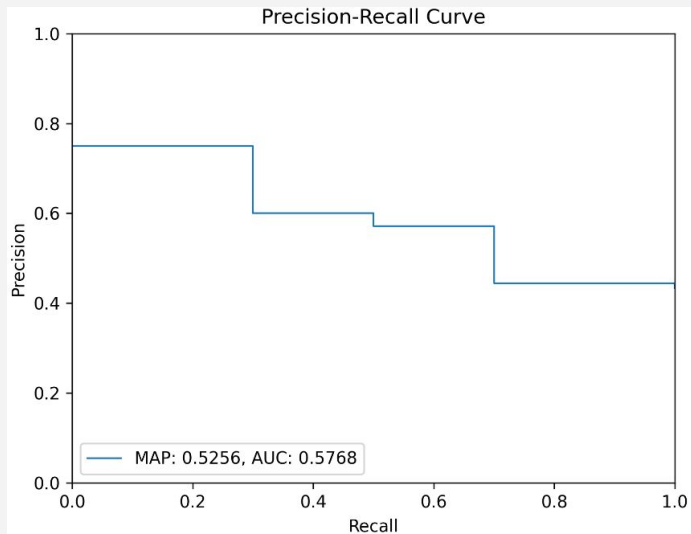
Rank System	Simple	Advanced	Improved	Semantic
P@10	0.1000	0.4000	0.6000	0.2000
P@30	0.2000	0.4667	0.4333	0.1000
AvP	0.1851	0.4216	0.5256	0.4907

Table 13: P@10, P@30, AvP for query 5

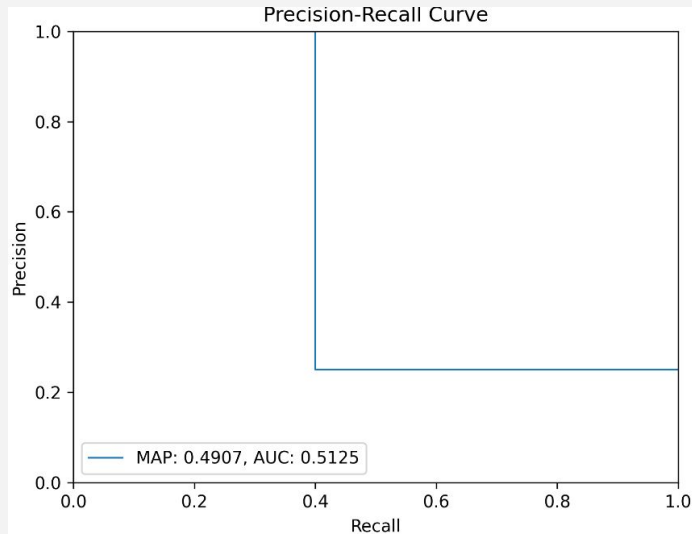
Query

Pop songs that talk about the past.

Query 5 - Improved Schema



Query 5- Semantic Schema



Queries

Final Conclusions

Global System	Simple	Advanced	Improved	Semantic
MAP	0.3848	0.7632	0.7869	0.3026

Future work

- Implementation with **multiple languages** for better search results
- **eDisMax** fine tuning
- **Interface Improvement** for a robust system addressing a potential user

A Year To Listen: Song Search System.

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